[FULL NAME]

AI / RAG & Backend Engineer

[Phone Number] | [Email Address] | Ho Chi Minh City, Vietnam | [GitHub / Portfolio Link]

EDUCATION

University of Transport Ho Chi Minh City (UTH) *2022 – 2026*

Bachelor of Information Technology (Coursework Completed; Graduation Pending) *Ho Chi Minh City, Vietnam*

• **GPA:** 3.34 / 4.0 • **Thesis:** ARQ-RAG — **Defense Score: 9.0 / 10.0**

CORE SKILLS

AI & RAG: Semantic Retrieval, Embeddings, Prompt Engineering, Output Evaluation, Query Rewriting, Citation Grounding, LangChain, QLoRA, Ragas

Vector Search: Qdrant, Supabase pgvector, IVF Indexing, Vector Quantization, Memory-Mapped I/O (mmap), Retrieval Evaluation

Backend & Cloud: JavaScript (Node.js, Express), Python (FastAPI), Java (Spring Boot), Docker Compose, Nginx, MongoDB, Redis, GCP Cloud Functions, Git

Languages: English: A2 (technical reading & documentation) • Vietnamese: Native

SELECTED PROJECTS & RESEARCH

ARQ-RAG — Adaptive Vector Retrieval Engine *2026*

First Author | Algorithmic Design & AI-Assisted Rust | Live Demo ***Thesis: 9.0 / 10.0***

- **My role:** Designed system architecture (IVF indexing, residual quantization, adaptive cluster routing). Generated low-level Rust engine (AVX2 SIMD, mmap) with AI coding agents; reviewed and verified all code.
- **RAM:** Cut active RAM from 1,931 MB (FAISS) to **102.6 MB (18.8×)** on 5M Wikipedia DPR vectors (D=768, single-thread CPU) by keeping only 4-bit codes in RAM; raw vectors served via zero-copy mmap.
- **Speed & Recall:** Achieved **4,522 QPS** (87.2% Recall@16 pre-rerank) and **99.2% Recall@16** after Float32 rerank at 3,450 QPS. Built 500K vector index in 20.2s (1.93× faster than FAISS IVFPQ).
- **Learned:** Vector quantization theory, memory hierarchy trade-offs (RAM vs. disk vs. cache), and reproducible benchmarks. Paper submitted to AMI 2026 *[Under Review]*.

EduChatbot — Vietnamese RAG Courseware Assistant *Jun 2026*

Project Lead (6-person team) | Python, FastAPI, Java 17, Spring Boot, Qdrant, Llama 3.1 8B | GitHub *Java Course Project*

- **RAG Pipeline & Integration:** Designed end-to-end RAG workflow (Vietnamese document parsing/chunking, 384D embeddings, Qdrant vector retrieval, citation verification to reduce hallucinations); built async FastAPI bridge integrating with Spring Boot services.
- **Evaluation, Cloud & Team Lead:** Led a 6-person team (sprint tracking, code reviews) and deployed endpoints on GCP Cloud Functions with Supabase + Qdrant Cloud; ran blind A/B Ragas evaluation showing RAG achieved higher faithfulness (0.303 vs. 0.292) and lower latency (2.00s vs. 3.75s) over fine-tuning.

Waterbase — Self-Hosted Multi-Tenant BaaS Platform *Nov 2025*

Sole Developer | Node.js, Express, MongoDB, Redis, Docker Compose, Nginx | Live Demo *Graduation Internship Project*

- **Backend Architecture & Ops:** Solely architected and deployed a modular 7-service microservices platform (Auth, App, WaterDB, Realtime, Storage, Rule Engine, Analytics) with Nginx API Gateway, running reliably in production on a 1 GB RAM / 1 vCPU VPS.
- **Multi-Tenancy & Access Control:** Implemented Database-per-Application isolation guaranteeing physical tenant separation, JWT auth with refresh rotation, and a declarative Rule Engine (RBAC) for collection-level permissions.

Internal AI Assistant — Lotus Hackathon VNG 2026 *Mar 2026*

Backend Developer (2-person team) | Node.js, Express, Supabase pgvector, Gemini API, n8n | Video Demo *VNG Corporation*

- **Rapid AI Backend (36h):** Built and delivered a working Node.js/Express backend in **36 hours**, featuring Gemini-powered RAG on Supabase pgvector, clientId-based multi-tenant isolation, and a 3-tier fallback intent router (parser → Gemini → keyword matching) with n8n automation.